

Hydro Model Architectures

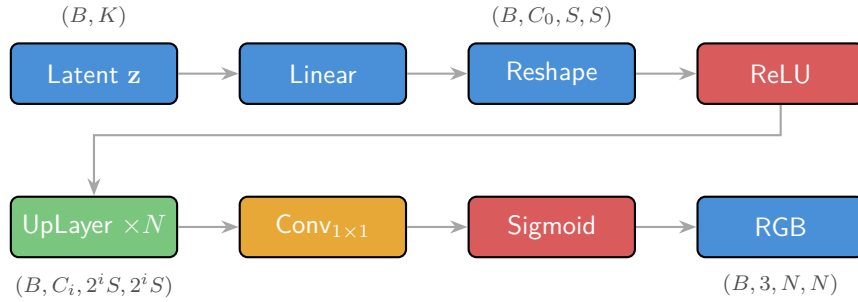
autopath

1 Overview

The **Hydro** datablock provides a flexible framework for latent-to-image decoders. It is designed to invert feature extractors (e.g., GigaPath) by mapping their embeddings back to the original image space. The system supports multiple architectural versions, dispatched via the **version** parameter in the model configuration.

2 Model_0: Deconvolutional Decoder

Model_0 is the baseline architecture, using progressive upsampling via transposed convolutions or bilinear interpolation to reconstruct $N \times N$ RGB images from K -dimensional latent vectors.



2.1 Architecture Details

2.1.1 Initial Projection

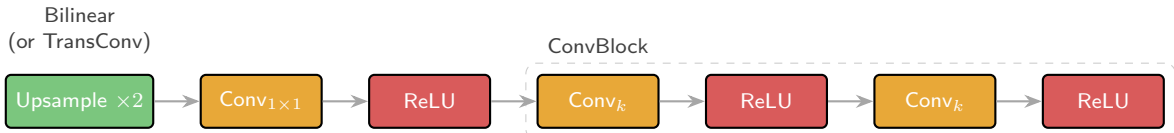
The latent vector $\mathbf{z} \in \mathbb{R}^K$ is projected to a spatial feature map:

$$\mathbf{z} \xrightarrow{\text{Linear}} \mathbb{R}^{C_0 \cdot S^2} \xrightarrow{\text{Reshape}} \mathbb{R}^{C_0 \times S \times S} \xrightarrow{\text{ReLU}} \mathbf{F}_0$$

where S is the **initial_size** (default 8) and $C_0 = H \cdot m_0$ with $H = \text{hidden_channels}$ and m_0 the first channel multiplier.

2.1.2 Progressive Upsampling

The number of upsample layers is $N_{\text{up}} = \log_2(N/S)$. Each **UpLayer** doubles spatial resolution:



Two upsampling modes are available, controlled by `use_bilinear_upsampling`:

- **Bilinear** (default): bilinear interpolation $\times 2$ followed by 1×1 conv + ReLU.
- **Transposed convolution**: `ConvTranspose2d` with 2×2 kernel and stride 2.

2.1.3 Output Head

A 1×1 convolution maps the final feature channels to 3 (RGB), followed by sigmoid activation and scaling:

$$\mathbf{I} = \text{sigmoid}(\text{Conv}_{1 \times 1}(\mathbf{F}_{N_{\text{up}}})) \times 255$$

2.2 Tensor Dimensions (Default Config)

With $K = 1536$, $N = 256$, $S = 8$, $H = 256$, multipliers $(8, 4, 2, 1, 1)$:

Stage	Channels	Spatial	Tensor Shape
Input latent	—	—	$(B, 1536)$
After projection	2048	8	$(B, 2048, 8, 8)$
UpLayer 0	2048	16	$(B, 2048, 16, 16)$
UpLayer 1	1024	32	$(B, 1024, 32, 32)$
UpLayer 2	512	64	$(B, 512, 64, 64)$
UpLayer 3	256	128	$(B, 256, 128, 128)$
UpLayer 4	256	256	$(B, 256, 256, 256)$
Final conv	3	256	$(B, 3, 256, 256)$

3 System Configuration

The `Hydro` datablock uses a shared configuration dataclass to control architecture and training parameters.

Parameter	Default	Description
<code>latent_dim</code>	1536	Input latent vector dimension K
<code>image_size</code>	256	Output resolution $N \times N$
<code>initial_size</code>	8	Spatial size before upsampling
<code>hidden_channels</code>	256	Base channel count H
<code>channel_multipliers</code>	$(8, 4, 2, 1, 1)$	Per-layer channel multiplier
<code>kernel_size</code>	3	Convolution kernel size
<code>use_batch_norm</code>	True	BatchNorm in conv blocks
<code>use_bilinear_upsampling</code>	True	Bilinear vs. transposed conv
<code>loss_type</code>	mse	MSE, L1, or L0 reconstruction loss
<code>version</code>	0	Model architecture version

4 Loss Functions

The following reconstruction losses are supported:

$$\mathcal{L} = \begin{cases} \frac{1}{3N^2} \|\hat{\mathbf{I}} - \mathbf{I}^*\|_2^2 & \text{if loss_type = "mse"} \\ \frac{1}{3N^2} \|\hat{\mathbf{I}} - \mathbf{I}^*\|_1 & \text{if loss_type = "l1"} \\ \frac{1}{3N^2} \sum \log(1 + \frac{(\hat{\mathbf{I}} - \mathbf{I}^*)^2}{\epsilon}) & \text{if loss_type = "l0"} \end{cases}$$

where $\epsilon = 10^{-3}$ and $\mathbf{I}^* \in [0, 255]^{3 \times N \times N}$.